

Literature Status: The Donoho–Stark Interval Conjectures

Run fa_banach_001 — lane 10

13 August 2026

Status: literature already answered. The continuous and circle-polynomial interval conjectures stated in arXiv:2307.04558 are both proved, without their former small-product restrictions, by Theorems 1.1 and 2.1 of the separate later paper arXiv:2607.19192.

1 Original questions

O. Baeza Guasch, *On an uncertainty result by Donoho and Stark*, arXiv:2307.04558 [1], restates the continuous conjecture as Conjecture 1 on arXiv PDF page 1. With $P_{\mathcal{T}}$ denoting multiplication by the indicator of a measurable time set and $P_{\mathcal{W}}$ the projection onto frequencies in an interval $\mathcal{W}$, it asks:

Source conjecture (continuous version). *Among measurable $\mathcal{T} \subset \mathbb{R}$ of fixed finite measure, the supremum of $\|P_{\mathcal{W}}P_{\mathcal{T}}\|$ is attained when $\mathcal{T}$ is an interval.*

The same paper formulates Conjecture 2 on arXiv PDF page 10. For analytic polynomials $p(z) = \sum_{k=0}^n a_k z^k$ on the unit circle and measurable $E \subset \mathbb{T}$ of prescribed angular measure, it asks:

Source conjecture (circle-polynomial version). *The maximum of*

$$\sup_{0 \neq p \in \mathcal{A}_n} \frac{\int_E |p(e^{it})|^2 dt}{\int_{\mathbb{T}} |p(e^{it})|^2 dt}$$

over all such E is attained when E is a circle interval.

Baeza Guasch proves the continuous statement only when $|\mathcal{W}||\mathcal{T}| \leq 1$ and the circle statement only when $n\delta \leq \pi$ for $|E| = 2\delta$.

2 Separate later answer

The separate later paper of L. D. Abreu and M. Speckbacher, arXiv:2607.19192 [2], proves both unrestricted statements.

Their concentration constant for a bounded interval Ω and a measurable finite-measure set E is

$$\mathfrak{C}_{\Omega}(E) = \sup_{0 \neq f \in PW(\Omega)} \frac{\int_E |f|^2}{\int_{\mathbb{R}} |f|^2} = \|P_{\Omega}M_E\|^2.$$

Theorem 1.1 on arXiv PDF page 1 states:

Later result (Theorem 1.1 of arXiv:2607.19192). *If $\Omega \subset \mathbb{R}$ is a bounded interval of positive length, $0 < |E| < \infty$, and I is any interval with $|I| = |E|$, then*

$$\mathfrak{C}_\Omega(E) \leq \mathfrak{C}_\Omega(I), \quad \|P_\Omega M_E\| \leq \|P_\Omega M_I\|.$$

This is exactly Conjecture 1 after identifying $M_E = P_{\mathcal{T}}$ and $P_\Omega = P_{\mathcal{W}}$. On PDF page 2 the supporting authors explicitly state that this confirms the Donoho–Stark conjecture, cite Baeza–Guasch’s product-1 result, and say that their theorem removes all restrictions on $|\Omega||E|$.

For the circle let $\mathcal{A}_N = \text{span}_{\mathbb{C}}\{1, e^{it}, \dots, e^{iNt}\}$ and $T_E^{(N)} = P_N M_E P_N|_{\mathcal{A}_N}$. Equation (5) of the later paper identifies

$$\lambda_{\max}(T_E^{(N)}) = \sup_{0 \neq p \in \mathcal{A}_N} \frac{\int_E |p(t)|^2 dt}{\int_{\mathbb{T}} |p(t)|^2 dt}.$$

Theorem 2.1 on arXiv PDF page 3 states:

Later result (Theorem 2.1 of arXiv:2607.19192). *For every $N \geq 0$, every measurable $E \subset \mathbb{T}$, and every circle interval I with $|I| = |E|$,*

$$\lambda_{\max}(T_E^{(N)}) \leq \lambda_{\max}(T_I^{(N)}).$$

This Rayleigh quotient and polynomial space are precisely those of source Conjecture 2. The later theorem has no $N\delta \leq \pi$ hypothesis, so it is the full circle-polynomial answer.

3 Scope and provenance

Both source conjectures are completely answered. The later results prove that intervals are maximizers; neither theorem asserts that all maximizers are intervals, but the source conjectures did not request uniqueness.

The continuous identification is explicit in the supporting paper. The circle identification is an exact equation-by-equation match; the supporting paper cites Baeza–Guasch, although it does not separately label Theorem 2.1 as the answer to Baeza–Guasch’s Conjecture 2. This packet records a later literature answer and claims no new mathematics.

A bounded search through 13 August 2026 checked the run indexes, exact title and conjecture phrases, official arXiv metadata, and the full source and PDF of arXiv:2607.19192.

References

- [1] O. Baeza Guasch, *On an uncertainty result by Donoho and Stark*, arXiv:2307.04558 (2023).
- [2] L. D. Abreu and M. Speckbacher, *Optimal concentration in the Paley–Wiener space*, arXiv:2607.19192 (2026).